CHENGHAO QIU

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EDUCATION

Texas A&M University
College Station, TX

PhD @ Computer Science | GPA: 4.0 / 4.0
Aug 2025 - Jun 2030 (Expected)

Tianjin University
Tianjin, China

Bachelor @ Computer Science | GPA: 3.75 / 4.0
Sep 2021 - Jun 2025

SKILLS AND INTERESTS

Program Language	Python | SQL | C++
ML Framework	PyTorch | Huggingface
Tools & Liberties	Git | Transformers | Scikit-learn | Pandas | Numpy

RESEARCH EXPERIENCE

ZHOU LAB, Texas A&M University

Graduate Research Assistant, with Prof. Yi Zhou — Aug 2025 - present

- Studying the reliability of **In-Context Learning (ICL)** through task preserving exemplar perturbations, revealing that correct demonstrations can still reduce downstream accuracy and formalizing this failure mode as contextual evidence shift across classification, reasoning, and math tasks.
- Developing **analog-friendly neural network** for efficient edge computing, including designing hardware-compatible NN architecture, developing noise-aware robustness training algorithms, and building a simulator to evaluate analog NN performance under realistic hardware conditions.

MAGICS LAB, Northwestern University

Undergraduate Research Assistant, with Prof. Han Liu — Apr 2024 - July 2025

- Proposed **GERM**, a fast and low-cost genomic foundation model for resource-constrained settings, by replacing standard attention with an outlier-free mechanism, improving fine-tuning performance by **37.98%** and quantization robustness by **64.34%** over the baseline.
- Developed the **GenoArmory** framework, the first unified benchmark for systematically evaluating the adversarial robustness of Genomic Foundation Models (GFMs), and introduced the **GenoAdv** dataset, achieving a **34.71%** increase in Defense Success Rate when used in training.

SELECTED PAPER

[1] Haozheng Luo*, **Chenghao Qiu***, Maojiang Su, Zhihan Zhou, Zoe Mehta, Guo Ye, Jerry Yao-Chieh Hu, Han Liu. Fast and Low-Cost Genomic Foundation Models via Outlier Removal. Forty-Second International Conference on Machine Learning (**ICML 2025**). link

MANUSCRIPT IN PREPARATION

[1] **Chenghao Qiu**, Chunli Peng, Yufeng Yang, Kuan-Hao Huang, Yi Zhou. When Correct Demonstrations Hurt: Rethinking the Role of Exemplars in In-Context Learning. Manuscript in preparation for submission to NIPS 2026. link

[2] Haozheng Luo*, **Chenghao Qiu***, Yimin Wang, Shang Wu, Jiahao Yu, Han Liu, Binghui Wang, Yan Chen. GenoArmory: A Unified Evaluation Framework for Adversarial Attacks on Genomic Foundation Models. Manuscript in preparation for submission to NIPS 2026 ED Track. link

*denotes equal contribution